

Hiram Kwak, Ph.D.

Postdoctoral Researcher

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CAREER

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- Sep.2025-present Post-doc researcher Pritzker School of Molecular Engineering, The University of Chicago, U.S.
— x-CT and PDF Characterization for all-solid-state batteries using a synchrotron beam source (PI: Prof. Shirley Meng)
- Oct. 2023-Sep.2025 Staff Engineer in All Solid Battery Cell Development Group, Samsung SDI Co. Ltd, Korea
— Participated in the scale-up of laboratory processes to pilot-scale production for all-solid-state batteries.
- Feb. 2022-Aug. 2023 Post-doc researcher in Department of Chemical and Biomolecular Engineering, Yonsei University, Korea
— Developed halide solid electrolytes and analyzed their structure and ionic transport mechanism for all-solid-state battery applications.

EDUCATION

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- Mar. 2018-Feb. 2022 **Ph. D** in Department of Energy Engineering, Hanyang University, Korea
Thesis: "Development of Sb-based Sulfide and Zr-based Halide Superionic Conductors for All-Solid-State Batteries", Advisor: Prof. Yoon Seok Jung
- Mar. 2013-Aug. 2017 **B.S.** in School of Energy and Chemical Engineering, Ulsan National Institute of Science and Technology (UNIST), Korea

PUBLICATIONS

First-Author

1. **Hiram Kwak**,† Kern Ho Park, Daseul Han, Kyung-Wan Nam, Hyungsub Kim*, Yoon Seok Jung*
"Li⁺ conduction in air-stable Sb-Substituted Li₄SnS₄ for all-solid-state Li-Ion batteries"
J.Power Sources, **2020**, 446, 227338.
2. **Hiram Kwak**,† Daseul Han,† Jeyne Lyoo, Juhyouon Park, Sung Hoo Jung, Yoonjae Han, Gihan Kwon, Hansu Kim, Seung-Tae Hong, Kyung-Wan Nam,* Yoon Seok Jung*
"New Cost-Effective Halide Solid Electrolytes for All-Solid-State Batteries: Mechanochemically Prepared Fe³⁺-substituted Li₂ZrCl₆"
Adv. Energy Mater. **2021**, 11, 2003190.
— Outstanding Papers Selected by the Korea Synchrotron Radiation User's Association
— Highlighted in '2021 Science Highlight' in National Synchrotron Light Source II (Brookhaven National Lab)
— Selected as an Inside Front Cover
3. **Hiram Kwak**,† Jeyne Lyoo,† Juhyouon Park, Yoonjae Han, Ryo Asakura, Arndt Remhof, Corsin Battaglia, Hansu Kim, Seung-Tae Hong,* Yoon Seok Jung*
"Na₂ZrCl₆ enabling highly stable 3 V all-solid-state Na-ion batteries"
Energy Storage Mater. **2021**, 37, 47
4. **Hiram Kwak**, Daseul Han, Jun Pyo Son, Jong Seok Kim, Juhyouon Park, Kyung-Wan Nam, Hyungsub Kim,* Yoon Seok Jung*
"Li⁺ conduction in aliovalent-substituted monoclinic Li₂ZrCl₆ for all-solid-state batteries: Li_{2+x}Zr_{1-x}M_xCl₆ (M = In, Sc)"
Chem. Eng. J. **2022**, 437, 135413.
5. **Hiram Kwak**,† Shuo Wang,† Juhyouon Park, Yunsheng Liu, Kyu Tae Kim, Yeji Choi, Yifei Mo,* Yoon Seok Jung*
"Emerging halide superionic conductors for all-solid-state batteries: Design, synthesis, and practical applications"
ACS Energy Lett. **2022**, 7, 1776.
— Selected to be featured as an ACS Editors' Choice
6. **Hiram Kwak**,† Jae-Seung Kim,† Daseul Han,† Jong Seok Kim, Ju Hyoun Park, Gihan Kwon, Seong-Min Bak, Unseon Heo, Changhyun Park, Hyun-Wook Lee, Kyung-Wan Nam,* Dong-Hwa Seo,* and Yoon Seok Jung*
"Boosting the interfacial superionic conduction of halide solid electrolytes for all-solid-state batteries"
Nat. commun. **2023**, 14, 2459.
7. **Hiram Kwak**,† Jong Seok Kim, Jae-Seung Kim, Juhyouon Park, Changhoon Kim, Dong-Hwa Seo, Kyung-Wan Nam, Yoon Seok Jung*
"Tuning the Properties of Halide Nanocomposite Solid Electrolytes with Diverse Oxides for All-Solid-State Batteries"
ACS Appl. Mater. Interfaces **2024**, 16, 49328.
8. Yeji Choi,a,† **Hiram Kwak**,† Jae-Seung Kim,† Daseul Han,† Jong Seok Kim, Juhyouon Park, Gihan Kwon, Kyung-Wan Nam,* Dong-Hwa Seo,* Yoon Seok Jung*
"Mechanism of Contrasting Ionic Conductivities in Li₂ZrCl₆ via I and Br Substitution"
Small **2025**, 21, e05926.

Co-Author

9. Young Eun Choi†, Kern Ho Park†, Dong Hyeon Kim, Dae Yang Oh, **Hiram Kwak**, Young-Gi Lee, Yoon Seok Jung*
"Coatable Li_4SnS_4 solid electrolytes prepared from aqueous solutions for all-solid-state lithium-ion batteries"
ChemSusChem **2017**, 10, 2605
10. Kern Ho Park, Dong Hyeon Kim, **Hiram Kwak**, Sung Hoo Jung, Hyun-Jae Lee, Abhik Banerjee, Jun Hee Lee, Yoon Seok Jung*
"Solution-Derived Glass-Ceramic $\text{NaI} \cdot \text{Na}_3\text{SbS}_4$ Superionic Conductors for All-Solid-State Na-Ion Batteries"
J. Mater. Chem. A **2018**, 6, 17192.
11. Chang Ki Moon, Hyun-Jae Lee, Kern Ho Park, **Hiram Kwak**, Jongwook W Heo, Keunsu Choi, Hyemi Yang, Maeng-Suk Kim, Seung-Tae Hong, Jun Hee Lee*, Yoon Seok Jung*
"Vacancy-Driven Na^+ Superionic Conduction in New Ca-doped Na_3PS_4 for All-Solid-State Na-ion Batteries"
ACS Energy Lett. **2018**, 3, 2504.
12. Dong Hyeon Kim,† Yong-Hyeok Lee,† Yong Bae Song, **Hiram Kwak**, Sang-Young Lee*, Yoon Seok Jung*
"Thin and Flexible Solid Electrolyte Membranes with Ultrahigh Thermal Stability Derived from Solution-Processable Li Argyrodites for All-Solid-State Li-Ion Batteries"
ACS Energy Lett. **2020**, 5, 718.
13. Saneyuki Ohno*, Tim Bernges, Johannes Buchheim, Marc Duchardt, Anna-Katharina Hatz, Marvin A. Kraft, **Hiram Kwak**, Aggunda L. Santhosha, Zhantao Liu, Nicolo Minafra, Fumika Tsuji, Atsushi Sakuda, Roman Schlem, Shan Xiong, Zhenggang Zhang, Philipp Adelhelm, Hailong Chen, Akitoshi Hayashi, Yoon Seok Jung, Bettina V. Lotsch, Bernhard Roling, Nella M. Vargas-Barbosa*, Wolfgang G. Zeier*
"How Certain Are the Reported Ionic Conductivities of Thiophosphate-Based Solid Electrolytes? An Interlaboratory Study"
ACS Energy Lett. **2020**, 5, 910.
14. Yong Bae Song, Dong Hyeon Kim, **Hiram Kwak**, Daseul Han, Sujin Kang, Jong Hoon Lee, Seongmin Bak, Kyung-Wan Nam, Hyun-Wook Lee*, Yoon Seok Jung*
"Tailoring Solution-Processable Li Argyrodites $\text{Li}_{6+x}\text{P}_{1-x}\text{M}_x\text{S}_5\text{I}$ (M = Ge, Sn) and Their Microstructural Evolution Revealed by Cryo-TEM for All-Solid-State Batteries"
Nano Lett. **2020**, 6, 4337.
15. Seunggoo Jun, Young Jin Nam, **Hiram Kwak**, Kyu Tae Lee, Dae Yang Oh, Yoon Seok Jung*
"Operando Differential Electrochemical Pressimetry for Probing Electrochemo-mechanics in All-Solid-State Batteries"
Adv. Funct. Mater. **2020**, 30, 2002535.
16. Yoonjae Han, Sung Hoo Jung, **Hiram Kwak**, Seunggoo Jun, Hunho Kwak, Jong Hoon Lee, Seung-Tae Hong, Yoon Seok Jung*
"Single- or Poly-crystalline Ni-Rich Layered Cathode, Sulfide or Halide Solid Electrolyte: Which will be the Winners for All-Solid-State Batteries?"
Adv. Energy Mater. **2021**, 11, 2100126.
17. Juhyoun Park†, Daseul Han†, **Hiram Kwak**, Yoonjae Han, Yong Jeong Choi, Kyung-Wan Nam*, Yoon Seok Jung*
"Heat Treatment Protocol for Modulating Ionic Conductivity via Structural Evolution of $\text{Li}_{3-x}\text{Yb}_{1-x}\text{M}_x\text{Cl}_6$ (M = Hf^{4+} , Zr^{4+}) New Halide Superionic Conductors for All-Solid-State Batteries"
Chem. Eng. J. **2021**, 425, 130630.
18. Yong Bae Song, **Hiram Kwak**, Woosuk Cho, Kyung Su Kim, Yoon Seok Jung*, Kern-Ho Park*
"Electrochemo-Mechanical Effects as a Critical Design Factor for All-Solid-State Batteries"
Curr. Opin. Solid State Mater. Sci. **2022**, 26, 100977.
19. Juhyoun Park†, Jun Pyo Son†, Wonseok Ko, Jae-Seung Kim, Yeji Choi, Hyungsub Kim, **Hiram Kwak**, Dong-Hwa Seo, Jongsoo Kim, Yoon Seok Jung*
" NaAlCl_4 : New Halide Solid Electrolyte for 3 V Stable Cost-Effective All-Solid-State Na-Ion Batteries"
ACS Energy Lett. **2022**, 7, 3293.
20. Mahander P. Singh, Se-Ho Kim*, Xuyang Zhou, **Hiram Kwak**, Alisson Kwiatkowski da Silva, Stoichko Antonov, Leonardo Shoji Aota, Changwon Jung, Yoon Seok Jung, Baptiste Gault*
"Near-Atomic-Scale Evolution of the Surface Chemistry in $\text{Li}[\text{Ni}, \text{Mn}, \text{Co}]\text{O}_2$ Cathode for Li-Ion Batteries Stored in Air"
Adv. Energy Sustainability Res. **2022**, 2200121.
21. Jong Seok Kim†, Seungwon Jung†, **Hiram Kwak**, Yoonjae Han, Suhwan Kim, Jongwoo Lim, Yong Min Lee*, Yoon Seok Jung*
"Synergistic Halide-Sulfide Solid Electrolytes for Ni-Rich Cathodes Design Guided by Digital Twin for All-Solid-State Li Batteries"
Energy Storage Mater. **2023**, 55, 193.
22. Jehoon Woo, Yong Bae Song, **Hiram Kwak**, Seunggoo Jun, Bo Yeong Jang, Juhyoun Park, Kyu Tae Kim, Changhyun Park, Chanhee Lee, Kern-Ho Park, Hyun-Wook Lee, Yoon Seok Jung*
"Liquid-Phase Synthesis of Highly Deformable and Air-Stable Sn-Substituted Li_3PS_4 for All-Solid-State Batteries Fabricated and Operated under Low Pressures"
Adv. Energy Mater. **2023**, 13, 2203292.
23. Yong Bae Song, Ki Heon Baeck, **Hiram Kwak**, Haechannara Lim, Yoon Seok Jung*
"Dimensional Strategies for Bridging the Research Gap between Lab-Scale and Potentially Practical All-Solid-State Batteries: The Role of Sulfide Solid Electrolyte Films"
Adv. Energy Mater. **2023**, 13, 2301142.
24. Boyeong Jang, Jehoon Woo, Yong Bae Song, **Hiram Kwak**, Juhyoun Park, Jong Seok Kim, Haechannara Lim, Yoon Seok Jung*
"Microwave heating enables near-carbonless liquid-phase-derived Li Argyrodites for all-solid-state batteries"
Energy Storage Mater. **2024**, 65, 103154.

25. Juhyouon Park†, Daseul Han†, Jun Pyo Son, **Hiram Kwak**, Wonseok Ko, Changhyun Park, Chanhee Lee, Hyun-Wook Lee, Jongsoo Kim, Kyung-Wan Nam*, Yoon Seok Jung*
"Extending the Electrochemical Window of Na⁺ Halide Nanocomposite Solid Electrolytes for 5 V-Class All-Solid-State Na-Ion Batteries"
ACS Energy Lett. **2024**, 9, 2222.
26. Changhoon Kim, Juhyouon Park, **Hiram Kwak**, Jae-Seung Kim, Seungwoo Jun, Dong-Hwa Seo, Yoon Seok Jung*
"KTaCl₆: High-Voltage Stable Potassium-Ion Conducting Chloride Solid Electrolyte"
Energy Storage Mater. **2024**, 71, 103618.

CONFERENCE

1. 2F23, The 59th Battery Symposium in Japan, Osaka International Convention Center, 5-3-51, Nakanoshima Kita-ku, Osaka, Japan, Nov. 27-29, 2018 (**On-site Oral, International**)
"Developments of room-temperature-operable all-solid-state Na-ion batteries using Na⁺ Superionic Conductors"
Hiram Kwak, Kern Ho Park, Yoon Seok Jung*
2. YONSEI-BUT International Workshop: Nanomaterials for Chemical Engineering, Feb. 2, 2021 (**Virtual Oral, International**)
"New Halide Superionic Conductors Using Earth-Abundant Elements For All-Solid-State Batteries"
Hiram Kwak, Yoon Seok Jung*
3. 2021 Virtual MRS Spring Meeting & Exhibit, Apr. 21, 2021 (**Virtual Oral, International**)
"Fe³⁺-Substituted Li₂ZrCl₆: New Halide Li⁺ Superionic Conductors for All-Solid-State Batteries"
Hiram Kwak, Da-Seul Han, Juhyouon Park, Yoonjae Han, Kyung-Wan Nam, Yoon Seok Jung*
4. 2021 Virtual MRS Fall Meeting & Exhibit, December 7th, 2021 (**Virtual Oral, International**)
"New Halide Na⁺ Conductors Using Earth Abundant Elements for All-Solid-State Na⁺ Batteries"
Hiram Kwak, Jeyne Lyoo, Juhyouon Park, Yoonjae Han, Seung-Tae Hong and Yoon Seok Jung*
5. 2022 MRS Spring Meeting & Exhibit, May 12th, 2022 (**On-site Oral, International**)
"Investigation of Li⁺ Migration in Monoclinic Li_{2+x}Zr_{1-x}M_xCl₆ (M = Sc, In)"
Hiram Kwak, Jong Seok Kim, Jun Pyo Son, Juhyouon Park, Yeji Choi, Daseul Han, Hyungsub Kim, Kyung-Wan Nam and Yoon Seok Jung*

Patents (Halide SE)

1. 리튬할라이드계 고체전해질, 이의 제조방법 및 상기 리튬할라이드계 고체전해질을 포함하는 전고체 전지 (Korea)
Hiram Kwak, Juhyouon Park and Yoon Seok Jung*
10-2657399
2. 나트륨할라이드계 고체전해질, 이의 제조방법 및 이를 포함하는 전고체전지 (Korea)
Hiram Kwak, and Yoon Seok Jung*
10-2364838
3. 리튬할라이드계 고체전해질, 이의 제조방법 및 이를 포함하는 전고체전지 (Korea)
Hiram Kwak, and Yoon Seok Jung*
10-2428607
4. 할라이드계 나노복합체, 이를 포함하는 고체 전해질, 이의 제조방법 및 상기 고체 전해질을 포함하는 전고체 전지(Korea)
Hiram Kwak, Juhyouon Park and Yoon Seok Jung*
10-2775492
5. 상온용 나트륨할라이드계 고체전해질의 제조방법 및 이에 따라 제조된 고체전해질
Hiram Kwak, Juhyouon Park and Yoon Seok Jung*
10-2838967

AWARDS

2019, 2021 Korea Electrochemical Society Spring Conference Outstanding Poster
2023 Samsung Global Technology Symposium Outstanding Poster Award